Actividad 6 - Integración numérica y solución de ecuaciones diferenciales

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Primer punto:

$$f'(1) = \frac{f(2) - f(0)}{2 - 0}$$

$$f'(1) = \frac{23 - 3}{2} = \frac{20}{2} = 10$$

Segundo punto:

$$f(x) = \begin{cases} x, & si\ 0\ \leq x\ \leq 0.5 \\ 0.5 - x\ , & si\ 0.5 < x\ \leq 1 \end{cases}$$

- Valor analítico:

$$\int_0^1 f(x)dx = \int_0^{0.5} x\ dx + \int_{0.5}^1 (0.5 - x)dx$$

$$\int x\ dx = \frac{x^2}{2} = \int_0^{0.5} x\ dx = \frac{(0.5)^2}{2} - \frac{0^2}{2} = \frac{0.25}{2} = 0.125$$

$$\int (0.5 - x)dx = 0.5x - \frac{x^2}{2} = \left[0.5x - \frac{x^2}{x}\right]_{0.5}^{1}$$

$$= \left(0.5 * 1 - \frac{1^2}{2}\right) - \left(0.5 * 0.5 - \frac{(0.5)^2}{2}\right)$$

$$= (0.5 - 0.5) - (0.25 - 0.125) = 0 - 0.125 = -0.125$$

$$\int_0^1 f(x)dx = 0.125 + (-0.125) = 0$$

- Aproximaciones numéricas:
  - ❖ Método trapezoidal (0,1)

$$\int_0^1 f(x)dx = \frac{b - a}{2}(f(a) + f(b))$$

$$\int_0^1 f(x)dx = \frac{1 - 0}{2}\big(f(0) + f(1)\big) = \frac{1}{2}(0 + 0) = 0$$

- Método trapezoidal en (0, 0.5) y (0.5, 1)

$$\int_0^{0.5} f(x)dx = \frac{0.5-0}{2}\big(f(0)+f(0.5)\big) = \frac{0.5}{2}(0+0.5) = 0.125$$

$$\int_{0.5}^{1} f(x)dx = \frac{1-0.5}{2}\big(f(0.5)+f(1)\big) = \frac{0.5}{2}(0.5+0) = 0.125$$

$$\int_{0.5}^{1} f(x)dx = 0.125 + 0.125 = 0.25$$

- Método de Simpson 1/3 en (0, 1)

$$\int_0^1 f(x)dx = \frac{1-0}{6}\big(f(0)+4f(0.5)+f(1)\big) = \frac{1}{6}(0+4*0.5+0) = \frac{1}{6}*2 = 0.333$$

Tercer Punto:

$$\int_{-\pi}^{\pi} \frac{sin(x)}{x}dx$$

$$\frac{lim}{x \to 0} \qquad \frac{sin(x)}{x} = 1$$

Formula de Romberg:

$$I_2{}^{(2)} = \frac{4 * I\left(\frac{h}{2}\right) - I(h)}{3}$$

$$I(h) = \frac{\pi-(-\pi)}{2}\big(f(-\pi)+f(\pi)\big) = \pi * (f(-\pi)+f(\pi))$$

$$I(h/2) = \frac{\pi}{4}(f(-\pi)+2f(0)+f(\pi))$$

$$I_2{}^{(2)} = 4.1888$$

Quinto Punto:

$$I_1(t+h) = I_1(t) + h * \frac{dI_1}{dt}\big(I_1(t),\ I_2(t)\big)$$

$$I_2(t+h) = I_2(t) + h * \frac{dI_2}{dt}\big(I_1(t),\ I_2(t)\big)$$

$$\frac{dI_1}{dt} = (-4I_1 + 3I_2 + 6)$$

$$\frac{dI_2}{dt} = (-2.4I_1 + 1.6I_2 + 3.6)$$

$$t = 0,\ I_1 = 0,\ I_2 = 0$$

$$\frac{dI_1}{dt} = -4(0) + 3(0) + 6 = 6$$

$$\frac{dI_2}{dt} = -2.4(0) + 1.6(0) + 3.6 = 3.6$$

$$I_1(0.1) = I_1(0) + 0.1 * 6 = 0 + 0.6 = 0.6$$

$$I_2(0.1) = I_2(0) + 0.1 * 3.6 = 0 + 0.36 = 0.36$$

$$t = 0.1\ a\ t = 0.2$$

$$\frac{dI_1}{dt} = -4(0.6) + 3(0.36) + 6 = -2.4 + 1.08 + 6 = 4.68$$

$$\frac{dI_2}{dt} = -2.4(0.6) + 1.6(0.36) + 3.6 = -1.44 + 0.576 + 3.6 = 2.736$$

$$I_1(0.2) = I_1(0.1) + 0.1 * 4.68 = 0.6 + 0.468 = 1.068$$

$$I_2(0.2) = I_2(0.1) + 0.1 * 2.736 = 0.36 + 0.2736 = 0.6336$$

$h = 0.05$:

$I_1(3) = 2.0413$

$I_2(3) = 0.6525$

$h = 0.1$:

$I_1(3) = 2.0468$

$I_2(3) = 0.6584$

$h = 0.5$:

$I_1(3) = 1.9915$

$I_2(3) = 0.5898$

$h = 1.0$:

$I_1(3) = 5.2800$

$I_2(3) = 2.7360$